

4

LAPLACE TRANSFORM

LAPLACE TRANSFORM

The Laplace transform changes a function into a new function by using a process that involves integration. The Laplace transform is named after mathematician and astronomer De Laplace who used a similar transform called Z-transform in his work on probability theory. The powerful practical Laplace transform techniques were developed by the English electrical engineer Oliver Heaviside and were often called Heaviside Calculus.

An integral transform is any transform of the form

$$F(s) = \int k(s, t)f(t)dt$$

The input of this transform is a function f and the output is another function F . There are numerous useful integral transforms. Each is specified by a choice of the function k of two variables, the kernel function, integral kernel or nucleus of the transform. Some kernels have an associated inverse kernel $k^{-1}(s, t)$ which yields an inverse transform

$$f(t) = \int k^{-1}(s, t)F(s)ds$$

Two important Integral transforms are Laplace transform and Fourier transform. The detailed discussion of Fourier transform is in the next chapter.

4.1 LAPLACE TRANSFORM

If $f(t)$ is a function defined for all $t \geq 0$, its Laplace transform is the integral of $f(t)$ times e^{-st} from $t=0$ to $t=\infty$. It is a function of s denoted by $L(f)$ and $F(s)$

$$L(f) = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

The Laplace transform is a linear operation that is for any function $f(t)$ and $g(t)$ whose transform exist and any constants a and b the transform of $af(t)+bg(t)$ exists, and

$$L[af(t) + bg(t)] = aL(f(t)) + bL(g(t))$$

4.2 BASIC LAPLACE TRANSFORMS

1. $f(t) = 1, t \geq 0$

$$\begin{aligned} L(f) = L(1) &= \int_0^{\infty} e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} \\ &= \frac{1}{s}, \quad s > 0 \end{aligned}$$

2. $f(t) = e^{at}, t \geq 0; a$ is a constant

$$\begin{aligned} L(e^{at}) &= \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt = \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} \\ &= \frac{1}{s-a}, \quad s-a > 0 \end{aligned}$$

3. $f(t) = \sin wt$

$$L(\sin wt) = \int_0^{\infty} e^{-st} \sin wt dt$$

$$\left[\int e^{ax} \sin bxdx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) \right]$$

$$L(\sin wt) = \left[\frac{e^{-st}}{s^2 + w^2} (-s \sin wt - w \cos wt) \right]_0^{\infty} = \frac{w}{s^2 + w^2}$$

$$L_s = L(\sin wt) = \frac{w}{s^2 + w^2}$$

	$f(t)$	$F(s)=L(f(t))$
1	1	$\frac{1}{s}$
2	t	$\frac{1}{s^2}$
3	t^2	$\frac{2!}{s^3}$
4	t^n	$\frac{n!}{s^{n+1}}$
5	t^a	$\frac{\Gamma(a+1)}{s^{a+1}}$
6	e^{at}	$\frac{1}{s-a}$
7	$\sin wt$	$\frac{w}{s^2 + w^2}$
8	$\cos wt$	$\frac{s}{s^2 + w^2}$
9	$\sinh at$	$\frac{a}{s^2 - a^2}$
10	$\cosh at$	$\frac{s}{s^2 - a^2}$
11	$e^{at} \sin wt$	$\frac{w}{(s-a)^2 + w^2}$
12	$e^{at} \cos wt$	$\frac{s-a}{(s-a)^2 + w^2}$

4.3 SOLVED PROBLEMS

Find the Laplace transform of the following.

1. $2t + 8$

$$L(2t + 8) = 2L(t) + 8L(1) = \frac{2}{s^2} + \frac{8}{s}$$

2. $(a - bt)^2$

$$\begin{aligned} L[(a - bt)^2] &= L(a^2 - 2abt + b^2t^2) \\ &= a^2L(1) - 2abL(t) + b^2L(t^2) \\ &= \frac{a^2}{s} - \frac{2ab}{s^2} + \frac{2b^2}{s^3} \end{aligned}$$

3. $\cos 2\pi t$

$$L(\cos 2\pi t) = \frac{s}{s^2 + 4\pi^2}$$

4. $\cos^2 wt$

$$\begin{aligned} L(\cos^2 wt) &= L\left(\frac{1 + \cos 2wt}{2}\right) \\ &= \frac{1}{2}[L(1) + L(\cos 2wt)] \\ &= \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2 + 4w^2}\right] = \frac{s^2 + 2w^2}{s(s^2 + 4w^2)} \end{aligned}$$

5. $e^{3t} \sinh t$

$$\begin{aligned} L(e^{3t} \sinh t) &= L\left[e^{3t} \left(\frac{e^t + e^{-t}}{2}\right)\right] \\ &= \frac{1}{2}[L(e^{4t}) + L(e^{2t})] \\ &= \frac{1}{2}\left(\frac{1}{s-4} + \frac{1}{s-2}\right) = \frac{s-3}{s^2 - 6s + 8} \end{aligned}$$

6. $e^{-t} \sinh 4t$

$$\begin{aligned} L(e^{-t} \sinh 4t) &= L\left(e^{-t} \left(\frac{e^{4t} + e^{-4t}}{2}\right)\right) \\ L(e^{-t} \sinh 4t) &= L\left(e^{-t} \left(\frac{e^{4t} + e^{-4t}}{2}\right)\right) \\ &= \frac{1}{2} (L(e^{3t}) + L(e^{-5t})) \\ &= \frac{1}{2} \left(\frac{1}{s-3} + \frac{1}{s+5}\right) = \left(\frac{s+1}{s^2+2s-15}\right) \end{aligned}$$

7. $\cos(wt + \theta)$

[KTU Dec 2018]

$$\begin{aligned} L(\cos(wt + \theta)) &= L(\cos wt \cos \theta - \sin wt \sin \theta) \\ &= \cos \theta L(\cos wt) - \sin \theta L(\sin wt) \\ &= \cos \theta \frac{s}{s^2 + w^2} - \sin \theta \frac{w}{s^2 + w^2} = \frac{s \cos \theta - w \sin \theta}{s^2 + w^2} \end{aligned}$$

8. $1.5 \sin\left(3t - \frac{\pi}{2}\right)$

$$\begin{aligned} L\left[1.5 \sin\left(3t - \frac{\pi}{2}\right)\right] &= L\left[1.5 \left(\sin 3t \cos \frac{\pi}{2} - \cos 3t \sin \frac{\pi}{2}\right)\right] \\ &= -1.5 L(\cos 3t) = \frac{-1.5s}{s^2 + 9} \end{aligned}$$

9. $(t-1)^3$

[KTU July 2018]

$$L((t-1)^3) = L(t^3 - 3t^2 - 3t - 1) = \frac{6}{s^4} - \frac{6}{s^3} - \frac{3}{s^2} - \frac{1}{s}$$

10. $\sin 3t \cos 2t$

[KTU Apr 2018]

$$\begin{aligned} L(\sin 3t \cos 2t) &= \frac{1}{2} L(2 \sin 3t \cos 2t) \\ &= \frac{1}{2} L(\sin 5t + \sin t) \\ &= \frac{1}{2} \left[\frac{5}{s^2 + 25} + \frac{1}{s^2 + 1} \right] \end{aligned}$$

11. $\cos^3 2t$

$$\cos^3 2t = \frac{1}{4} \cos 6t + \frac{3}{4} \cos 2t$$

$$L(\cos^3 2t) = \frac{1}{4} L(\cos 6t) + \frac{3}{4} L(\cos 2t)$$

$$= \frac{1}{4} \frac{s}{s^2 + 36} + \frac{3}{4} \frac{s}{s^2 + 4}$$

$$= \frac{s(s^2 + 28)}{(s^2 + 36)(s^2 + 4)}$$

12. $3e^{5t} + (t+2)^2 + 2\cos 3t$

$$L[3e^{5t} + (t+2)^2 + 2\cos 3t]$$

$$= 3L(e^{5t}) + L(t^2) + 4L(t) + 4L(1) + 2L(\cos 3t)$$

$$= \frac{3}{s-5} + \frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s} + \frac{2s}{s^2 + 9}$$

4.4 FIRST SHIFTING THEOREM

If $f(t)$ has the transform $F(s)$ where $s > k$ for some k , then $e^{at}f(t)$ has the transform $F(s-a)$.

$$L(e^{at}f(t)) = F(s-a)$$

Then,

$$e^{at}f(t) = L^{-1}(F(s-a))$$

Now, we have

$$L(e^{at} \cos wt) = \frac{s-a}{(s-a)^2 + w^2}$$

$$L(e^{at} \sin wt) = \frac{w}{(s-a)^2 + w^2}$$

4.5 SOLVED PROBLEMS

I Find the Laplace transform of the following functions.

1. $t^3 e^{-2t}$

$$L(e^{-2t} t^3) = [L(t^3)]_{s \rightarrow s+2} = \left[\frac{3!}{s^4} \right]_{s \rightarrow s+2} = \frac{6}{(s+2)^4}$$

2. $2e^{-\frac{1}{2}t} \sin 4\pi t$

$$\begin{aligned} L(2e^{-\frac{1}{2}t} \sin 4\pi t) &= 2[L(\sin 4\pi t)]_{s \rightarrow s+\frac{1}{2}} \\ &= 2 \left[\frac{4\pi}{s^2 + 16\pi^2} \right]_{s \rightarrow s+\frac{1}{2}} = \frac{8\pi}{\left(s + \frac{1}{2}\right)^2 + 16\pi^2} \\ &= \frac{32\pi}{(2s+1)^2 + 64\pi^2} \end{aligned}$$

3. **Sinh t cost**

[KTU July 2017]

$$\begin{aligned} L(\sinh t \cos t) &= L\left(\left(\frac{e^t - e^{-t}}{2}\right) \cos t\right) \\ &= \frac{1}{2} [L(e^t \cos t) - L(e^{-t} \cos t)] \\ &= \frac{1}{2} \{ [L(\cos t)]_{s \rightarrow s-1} - [L(\cos t)]_{s \rightarrow s+1} \} \\ &= \frac{1}{2} \left[\left(\frac{s}{s^2 + 1} \right)_{s \rightarrow s-1} - \left(\frac{s}{s^2 + 1} \right)_{s \rightarrow s+1} \right] \\ &= \frac{1}{2} \left[\frac{s-1}{(s-1)^2 + 1} - \frac{s+1}{(s+1)^2 + 1} \right] \\ &= \frac{1}{2} \left[\frac{s-1}{s^2 - 2s + 2} - \frac{s+1}{s^2 + 2s + 2} \right] \end{aligned}$$

4. $4te^{-2t}$

[KTU May 2017]

$$L(4te^{-2t}) = 4L(e^{-2t}t) = 4[L(t)]_{s \rightarrow s+2} = 4\left[\frac{1}{s^2}\right]_{s \rightarrow s+2} = \frac{4}{(s+2)^2}$$

5. $e^{-2t} \cos^2 t$

[KTU Apr 2018]

$$\begin{aligned} L(e^{-2t} \cos^2 t) &= [L(\cos^2 t)]_{s \rightarrow s+2} = \frac{1}{2}[L(1) + L(\cos 2t)]_{s \rightarrow s+2} \\ &= \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2 + 4}\right]_{s \rightarrow s+2} = \frac{1}{2}\left[\frac{1}{s+2} + \frac{s+2}{s^2 + 4s + 8}\right] \end{aligned}$$

6. te^{kt}

[KTU Dec 2018]

$$L(te^{kt}) = [L(t)]_{s \rightarrow s-k} = \left[\frac{1}{s^2}\right]_{s \rightarrow s-k} = \frac{1}{(s-k)^2}$$

7. $e^{-2t} [\cos 4t + 3 \sin 4t]$

$$L(\cos 4t + 3 \sin 4t) = \left[\frac{s}{s^2 + 16} + \frac{12}{s^2 + 16}\right]$$

$$L(e^{-2t} [\cos 4t + 3 \sin 4t]) = \left[\frac{s+12}{s^2 + 16}\right]_{s \rightarrow s+2} = \frac{s+14}{(s+2)^2 + 16}$$

8. $(t+1)^2 e^t$

$$L((t+1)^2) = L(t^2) + 2L(t) + L(1) = \frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} = \frac{2 + 2s + s^2}{s^3}$$

$$L((t+1)^2 e^t) = \frac{2 + 2(s-1) + (s-1)^2}{(s-1)^3} = \frac{s^2 + 1}{(s-1)^3}$$

II Find the inverse Laplace transform of the following functions.

1. $\frac{5s+1}{s^2-25}$

$$L^{-1}\left(\frac{5s+1}{s^2-25}\right) = 5L^{-1}\left(\frac{s}{s^2-25}\right) + L^{-1}\left(\frac{1}{s^2-25}\right)$$

$$= 5 \cosh 5t + \frac{\sinh 5t}{5}$$

$$2. \quad \frac{s}{L^2 s^2 + \frac{1}{4} \pi^2}$$

$$L^{-1} \left(\frac{s}{L^2 s^2 + \frac{\pi^2}{4}} \right) = \frac{1}{L^2} L^{-1} \left(\frac{s}{s^2 + \frac{\pi^2}{4L^2}} \right) = \frac{1}{L^2} \cos \left(\frac{\pi}{2L} t \right)$$

$$3. \quad \frac{1}{(s + \sqrt{2})(s - \sqrt{3})}$$

[KTU July 2017]

Using partial fractions,

$$\frac{1}{(s + \sqrt{2})(s - \sqrt{3})} = \frac{A}{s + \sqrt{2}} + \frac{B}{s - \sqrt{3}}$$

$$1 = A(s - \sqrt{3}) + B(s + \sqrt{2})$$

$$A = \frac{-1}{\sqrt{2} + \sqrt{3}}$$

$$B = \frac{1}{\sqrt{2} + \sqrt{3}}$$

$$\frac{1}{(s + \sqrt{2})(s - \sqrt{3})} = \frac{1}{\sqrt{2} + \sqrt{3}} \frac{1}{s - \sqrt{3}} - \frac{1}{\sqrt{2} + \sqrt{3}} \frac{1}{s + \sqrt{2}}$$

$$L^{-1} \left[\frac{1}{(s + \sqrt{2})(s - \sqrt{3})} \right]$$

$$= \frac{1}{\sqrt{2} + \sqrt{3}} \left[L^{-1} \left(\frac{1}{s - \sqrt{3}} \right) - L^{-1} \left(\frac{1}{s + \sqrt{2}} \right) \right] = \frac{1}{\sqrt{2} + \sqrt{3}} \left[e^{\sqrt{3}t} - e^{-\sqrt{2}t} \right]$$

$$4. \quad \frac{2}{s^4} - \frac{48}{s^6}$$

$$L^{-1} \left(\frac{2}{s^4} - \frac{48}{s^6} \right) = 2L^{-1} \left(\frac{1}{s^4} \right) - 48L^{-1} \left(\frac{1}{s^6} \right)$$

$$= 2 \frac{t^3}{3!} - 48 \frac{t^5}{5!} = \frac{t^3}{3} - \frac{2t^5}{5}$$

5. $\frac{-s+11}{s^2-2s-3}$

$$\begin{aligned} L^{-1}\left(\frac{-s+1}{s^2-2s-3}\right) &= L^{-1}\left(\frac{-(s-1)+10}{(s-1)^2-4}\right) \\ &= -L^{-1}\left(\frac{s-1}{(s-1)^2-4}\right) + 10L^{-1}\left(\frac{1}{(s-1)^2-4}\right) \\ &= -e^t L^{-1}\left(\frac{s}{s^2-4}\right) + 10e^t L^{-1}\left(\frac{1}{s^2-4}\right) \\ &= -e^t \cosh 2t + \frac{10}{2} e^t \sinh 2t \\ &= -e^t \left[\frac{e^{2t} + e^{-2t}}{2}\right] + 5e^t \left[\frac{e^{2t} - e^{-2t}}{2}\right] \\ &= 2e^{3t} - 3e^{-t} \end{aligned}$$

6. $\frac{2\pi}{(s+\pi)^3}$

$$L^{-1}\left(\frac{2\pi}{(s+\pi)^3}\right) = 2\pi e^{-\pi t} L^{-1}\left(\frac{1}{s^3}\right) = 2\pi e^{-\pi t} \frac{t^2}{2} = \pi t^2 e^{-\pi t}$$

7. $\frac{\pi}{s^2+4s\pi+3\pi^2}$

$$\begin{aligned} L^{-1}\left(\frac{\pi}{s^2+4s\pi+3\pi^2}\right) &= L^{-1}\left(\frac{\pi}{(s+2\pi)^2-\pi^2}\right) \\ &= e^{-2\pi t} L^{-1}\left(\frac{\pi}{s^2-\pi^2}\right) = e^{-2\pi t} \sinh \pi t \end{aligned}$$

$$8. \quad \frac{a_0}{s+1} + \frac{a_1}{(s+1)^2} + \frac{a_2}{(s+1)^3}$$

$$L^{-1} \left(\frac{a_0}{s+1} + \frac{a_1}{(s+1)^2} + \frac{a_2}{(s+1)^3} \right) = a_0 e^{-t} + a_1 t e^{-t} + a_2 e^{-t} \frac{t^2}{2}$$

$$9. \quad \frac{6s+7}{2s^2+4s+10}$$

$$L^{-1} \left(\frac{6s+7}{2s^2+4s+10} \right) = L^{-1} \left(\frac{6(s+1)+1}{2(s+1)^2+8} \right)$$

$$= \frac{1}{2} \left[6L^{-1} \frac{(s+1)}{(s+1)^2+4} + L^{-1} \frac{1}{(s+1)^2+4} \right]$$

$$= \frac{1}{2} \left[6e^{-t} L^{-1} \frac{s}{(s^2+4)} + e^{-t} L^{-1} \frac{1}{(s^2+4)} \right]$$

$$= \frac{1}{2} \left[6e^{-t} \cos 2t + \frac{e^{-t}}{2} \sin 2t \right]$$

$$= 3e^{-t} \cos 2t + \frac{e^{-t}}{4} \sin 2t$$

$$10. \quad \frac{2s+1}{s^2+2s+5}$$

[KTU May 2017]

$$L^{-1} \left[\frac{2s+1}{s^2+2s+5} \right] = L^{-1} \left[\frac{2(s+1)-1}{(s+1)^2+4} \right]$$

$$= 2L^{-1} \left[\frac{s+1}{(s+1)^2+4} \right] - L^{-1} \left[\frac{1}{(s+1)^2+4} \right]$$

$$= 2e^{-t} \cos 2t - e^{-t} \frac{\sin 2t}{2}$$

11. $\frac{s-4}{s^2-4}$

[KTU Apr 2018]

$$\begin{aligned} L^{-1}\left[\frac{s-4}{s^2-4}\right] &= L^{-1}\left[\frac{s}{s^2-4}\right] - 4L^{-1}\left[\frac{1}{s^2-4}\right] \\ &= \cosh 2t - 2\sinh 2t \end{aligned}$$

12. $\frac{4}{s^2-2s-3}$

$$L^{-1}\left[\frac{4}{s^2-2s-3}\right] = 4L^{-1}\left[\frac{1}{(s-1)^2-4}\right] = 2e^t \sinh 2t$$

13. $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$

$$\frac{2s^2-6s+5}{(s-1)(s-2)(s-3)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$$

$$2s^2-6s+5 = A(s-2)(s-3) + B(s-1)(s-3) + C(s-1)(s-2)$$

$$s=1, 4=2A \Rightarrow A=1/2$$

$$s=2, 1=-B \Rightarrow B=-1$$

$$s=3, 5=2C \Rightarrow C=5/2$$

$$\frac{2s^2-6s+5}{(s-1)(s-2)(s-3)} = \frac{1/2}{s-1} + \frac{-1}{s-2} + \frac{5/2}{s-3}$$

$$L^{-1}\left[\frac{2s^2-6s+5}{(s-1)(s-2)(s-3)}\right] = \frac{1}{2}L^{-1}\left[\frac{1}{s-1}\right] - L^{-1}\left[\frac{1}{s-2}\right] + \frac{5}{2}L^{-1}\left[\frac{1}{s-3}\right]$$

$$= \frac{1}{2}e^t - e^{2t} + \frac{5}{2}e^{3t}$$

14. $\frac{4s+5}{(s+1)^2(s-2)}$

$$\frac{4s+5}{(s+1)^2(s-2)} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s-2}$$

ANSWERS

$$1. \quad \frac{2}{s^3} - \frac{2}{s^2}$$

$$3. \quad \frac{s \cos \theta + w \sin \theta}{s^2 + w^2}$$

$$5. \quad \frac{k(s+a)}{(s+a)^2 + w^2}$$

$$7. \quad \frac{18}{s(s^2 + 36)}$$

$$9. \quad \frac{1}{s+4} - \frac{12}{s^3} + \frac{8}{s^2+4}$$

$$11. \quad \frac{1}{2} \left[\frac{s}{(s-a)^2 + a^2} - \frac{a}{(s+a)^2 + a^2} \right]$$

$$13. \quad \frac{10}{s(s-4)}$$

II.

$$1. \quad 0.2 \cos(1.4t) + \sin(1.4t)$$

$$2. \quad 4 \cosh(4t) + 8 \sinh(4t)$$

$$3. \quad \frac{1}{b-a} e^{-at} + \frac{1}{a-b} e^{-bt}$$

$$4. \quad 3e^{-t} t^2$$

$$5. \quad \frac{3}{2} e^{-\sqrt{3}t} t^5$$

$$6. \quad 2e^t \sinh 2t$$

$$7. \quad e^{-kt} (a \cos \pi t + b \sin \pi t)$$

$$8. \quad k_0 + k_1 e^{at} t$$

$$2. \quad \frac{s}{s^2 + 4\pi^2}$$

$$4. \quad \frac{1}{s^2 + 4}$$

$$6. \quad \frac{3(s^2 + 5)}{(s^2 + 2)(s^2 + 25)}$$

$$8. \quad \frac{48}{(s^2 - 4)(s^2 - 36)}$$

$$10. \quad \frac{a \cos b + s \sin b}{s^2 + a^2}$$

$$12. \quad \frac{s-2}{s^2 - 2s - 8}$$

9. $1 + 2t + \frac{5}{2}t^2$

10. $\frac{7}{6}e^t + \frac{5}{6}e^{-5t}$

11. $2e^{-2t} \cos 3t + e^{-2t} \frac{\sin 3t}{3}$

12. $e^t - e^{-t} \cos 2t + 3e^{-t} \frac{\sin 2t}{2}$

13. $3e^t (\cos t + \sin t)$

14. $e^{-t} \left(2 \cos 2t - \frac{\sin 2t}{2} \right)$

4.6 LAPLACE TRANSFORM OF DERIVATIVES

If $f(t)$ is continuous for all $t > 0$ and $f'(t)$ is piecewise continuous on every finite interval on the semi axis $t > 0$ the Laplace transform of the first derivative of $f(t)$ is

$$L(f'(t)) = sL(f(t)) - f(0)$$

If $f(t)$ and $f'(t)$ are continuous for all $t > 0$ and $f''(t)$ is piecewise continuous on every finite interval on the semi axis $t > 0$, the Laplace transform of the second derivative of $f(t)$ is

$$L(f''(t)) = s^2L(f(t)) - sf(0) - f'(0)$$

Let $f, f', \dots, f^{(n-1)}$ be continuous for all $t > 0$ and $f^{(n)}$ be piecewise continuous on every finite interval on the semi axis $t > 0$. Then the Laplace transform of $f^{(n)}(t)$ is

$$L(f^{(n)}(t)) = s^n L(f(t)) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0)$$

The Laplace transform is used for solving ODEs and initial value problems. For solving the initial value problem, first consider the Laplace transform of derivatives.

4.7 SOLVED PROBLEMS

I. Using Laplace transform of derivatives, find the Laplace transform of the following functions.

1. $t \cos 4t$

Solution

$$f(t) = t \cos 4t \quad ; \quad f'(t) = -4t \sin 4t + \cos 4t$$

$$f''(t) = -16t \cos 4t - 4 \sin 4t - 4 \sin 4t$$

$$f''(t) = -16t \cos 4t - 8 \sin 4t$$

$$L[f''(t)] = s^2 Lf(t) - sf(0) - f'(0)$$

$$L(-16t \cos 4t - 8 \sin 4t) = s^2 L(t \cos 4t) - 1$$

$$-16L(t \cos 4t) - 8L(\sin 4t) = s^2 L(t \cos 4t) - 1$$

$$(s^2 + 16)L(t \cos 4t) = 1 - 8L(\sin 4t)$$

$$(s^2 + 16)L(t \cos 4t) = 1 - 8\left(\frac{4}{s^2 + 16}\right) = \frac{s^2 - 16}{s^2 + 16}$$

$$L(t \cos 4t) = \frac{s^2 - 16}{(s^2 + 16)^2}$$

2. te^{at}

Solution

$$f(t) = te^{at}$$

$$f'(t) = tae^{at} + e^{at}$$

$$L(f'(t)) = sLf(t) - f(0)$$

$$L(tae^{at} + e^{at}) = sL(te^{at})$$

$$aL(te^{at}) + L(e^{at}) = sL(te^{at})$$

$$(s - a)L(te^{at}) = \frac{1}{s - a}$$

$$L(te^{at}) = \frac{1}{(s - a)^2}$$

II Using Laplace transform of derivatives, solve the initial value problems.

1. Solve the IVP $y'' + y' + 9y = 0, y(0) = 0.16, y'(0) = 0$

[KTU July 2017]

Solution

Take the Laplace transform

$$L(y'') + L(y') + L(9y) = 0$$

$$s^2 L(y) - sy(0) - y'(0) + sL(y) - y(0) + 9L(y) = 0$$

$$s^2 L(y) - 0.16s + sL(y) - 0.16 + 9L(y) = 0$$

$$(s^2 + s + 9)L(y) = 0.16(s + 1)$$

$$L(y) = \frac{0.16(s + 1)}{s^2 + s + 9}$$

$$L(y) = \frac{0.16\left(s + \frac{1}{2}\right) + 0.08}{\left(s + \frac{1}{2}\right)^2 + \frac{35}{4}}$$

$$y(t) = L^{-1} \left[\frac{0.16\left(s + \frac{1}{2}\right) + 0.08}{\left(s + \frac{1}{2}\right)^2 + \frac{35}{4}} \right]$$

$$= 0.16L^{-1} \left[\frac{\left(s + \frac{1}{2}\right)}{\left(s + \frac{1}{2}\right)^2 + \frac{35}{4}} \right] + 0.08L^{-1} \left[\frac{1}{\left(s + \frac{1}{2}\right)^2 + \frac{35}{4}} \right]$$

$$= 0.16e^{-\frac{1}{2}t} \cos \sqrt{\frac{35}{4}}t + \frac{0.08}{\sqrt{\frac{35}{4}}} e^{-\frac{1}{2}t} \sin \sqrt{\frac{35}{4}}t$$

$$= e^{-0.5t} (0.16 \cos 2.96t + 0.027 \sin 2.96t)$$

2. Solve the IVP $y' + \frac{2}{3}y = -4 \cos 2t, y(0) = 0$

Solution

$$L(y') + \frac{2}{3}L(y) = L(-4 \cos 2t)$$

$$sL(y) - y(0) + \frac{2}{3}L(y) = -4 \frac{s}{s^2 + 4}$$

$$\left(s + \frac{2}{3}\right)L(y) = \frac{-4s}{(s^2 + 4)\left(s + \frac{2}{3}\right)}$$

Using partial fractions

$$\frac{-4s}{(s^2 + 4)\left(s + \frac{2}{3}\right)} = \frac{As + B}{s^2 + 4} + \frac{C}{s + \frac{2}{3}}$$

$$-4s = (As + B)\left(s + \frac{2}{3}\right) + C(s^2 + 4)$$

Put $s = -\frac{2}{3}$

$$\frac{8}{3} = C\left(\frac{40}{9}\right) \Rightarrow C = \frac{3}{5}$$

Equating Coefficient of s^2

$$A + C = 0 \Rightarrow A = -\frac{3}{5}$$

Equating constant term

$$\frac{2}{3}B + 4C = 0$$

$$\frac{2}{3}B = \frac{-12}{5}$$

$$B = \frac{-2}{5}$$

$$L(y) = \frac{-\frac{3}{5}s - \frac{2}{5}}{s^2 + 4} + \frac{\frac{3}{5}}{s + \frac{2}{3}}$$

$$y(t) = -\frac{3}{5}L^{-1}\left(\frac{s}{s^2 + 4}\right) - \frac{2}{5}L^{-1}\left(\frac{1}{s^2 + 4}\right) + \frac{3}{5}L^{-1}\left(\frac{1}{s + \frac{2}{3}}\right)$$

$$= -\frac{3}{5}\cos 2t - \frac{2}{5} \cdot \frac{1}{2}\sin 2t + \frac{3}{5}e^{-\frac{2}{3}t}$$

$$= -\frac{3}{5}\cos 2t - \frac{1}{5}\sin 2t + \frac{3}{5}e^{-\frac{2}{3}t}$$

3. Solve the IVP $y'' - y = t$, $y(0) = 1$, $y'(0) = 1$

Solution

$$L(y'') - L(y) = L(t)$$

$$s^2L(y) - sy(0) - y'(0) - L(y) = \frac{1}{s^2}$$

$$(s^2 - 1)L(y) = \frac{1}{s^2} + s + 1$$

$$L(y) = \frac{1}{s^2(s^2-1)} + \frac{s+1}{s^2-1}$$

$$L(y) = \left(\frac{1}{s^2-1} - \frac{1}{s^2} \right) + \frac{1}{s-1}$$

$$y(t) = L^{-1}\left(\frac{1}{s^2-1}\right) - L^{-1}\left(\frac{1}{s^2}\right) + L^{-1}\left(\frac{1}{s-1}\right)$$

$$y(t) = \sinh t - t + e^t$$

4. Solve the IVP $y'' + y' - 6y = 0$, $y(0) = 1$, $y'(0) = 0$

[KTU Dec 2018]

Solution

$$L(y'') + L(y') - 6L(y) = 0$$

$$s^2L(y) - sy(0) - y'(0) + sL(y) - y(0) - 6L(y) = 0$$

$$(s^2 + s - 6)L(y) - s - 2 = 0$$

$$L(y) = \frac{s+2}{s^2+s-6}$$

Using Partial fractions

$$\frac{s+2}{s^2+s-6} = \frac{s+2}{(s+3)(s-2)} = \frac{A}{s+3} + \frac{B}{s-2}$$

$$s+2 = A(s-2) + B(s+3)$$

$$A = 1/5, B = 4/5$$

$$L(y) = \frac{1/5}{s+3} + \frac{4/5}{s-2}$$

$$y(t) = 1/5 L^{-1}\left(\frac{1}{s+3}\right) + 4/5 L^{-1}\left(\frac{1}{s-2}\right)$$

$$= \frac{1}{5} e^{-3t} + \frac{4}{5} e^{2t}$$

5. $y'' - \frac{1}{4}y = 0, y(0) = 12, y'(0) = 0$

Solution

$$L(y'') - \frac{1}{4}L(y) = 0$$

$$s^2L(y) - sy(0) - y'(0) - \frac{1}{4}L(y) = 0$$

$$\left(s^2 - \frac{1}{4}\right)L(y) = 12s$$

$$L(y) = \frac{12s}{s^2 - \frac{1}{4}}$$

$$y(t) = 12L^{-1}\left(\frac{s}{s^2 - \frac{1}{4}}\right)$$

$$y(t) = 12\cosh\left(\frac{t}{2}\right)$$

6. Solve $y'' - 3y' + 2y = 4t - 8, y(0) = 2, y'(0) = 7$

Solution

$$L(y'') - 3L(y') + 2L(y) = 4L(t) - 8L(1)$$

$$s^2L(y) - sy(0) - y'(0) - 3sL(y) + 3y(0) + 2L(y) = \frac{4}{s^2} - \frac{8}{s}$$

$$(s^2 - 3s + 2)L(y) - 2s - 7 + 6 = \frac{4}{s^2} - \frac{8}{s}$$

$$(s^2 - 3s + 2)L(y) = \frac{4}{s^2} - \frac{8}{s} + 2s + 1$$

$$L(y) = \frac{2s^3 + s^2 - 8s + 4}{s^2(s^2 - 3s + 2)}$$

Using partial fractions

$$\frac{2s^3 + s^2 - 8s + 4}{s^2(s^2 - 3s + 2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-2} + \frac{D}{s-1}$$

$$A = -1, B = 2, C = 2, D = 1$$

$$L(y) = \frac{-1}{s} + \frac{2}{s^2} + \frac{2}{s-2} + \frac{1}{s-1}$$

$$y(t) = -1 + 2t + 2e^{2t} + e^t$$

7. Solve $y'' + 2y' + 5y = 25t$, $y(0) = -2$, $y'(0) = -2$

[KTU May 2017]

Solution

$$[s^2 L(y) - sy(0) - y'(0)] + 2[sL(y) - y(0)] + 5L(y) = 25L(t)$$

$$(s^2 + 2s + 5)L(y) + 2s + 6 = \frac{25}{s^2}$$

$$(s^2 + 2s + 5)L(y) = \frac{25}{s^2} - 2s - 6 = \frac{25 - 2s^3 - 6s^2}{s^2}$$

$$L(y) = \frac{25 - 2s^3 - 6s^2}{s^2(s^2 + 2s + 5)}$$

$$\frac{25 - 2s^3 - 6s^2}{s^2(s^2 + 2s + 5)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs + D}{s^2 + 2s + 5}$$

$$25 - 2s^3 - 6s^2 = As(s^2 + 2s + 5) + B(s^2 + 2s + 5) + (Cs + D)s^2$$

$$A = -2, B = 5, C = 0, D = -7$$

$$L(y) = -\frac{2}{s} + \frac{5}{s^2} - \frac{7}{s^2 + 2s + 5}$$

$$y = -2 + 5t - 7 \frac{e^{-t} \sin 2t}{2}$$

8. Solve $y'' + y = 3 \cos 2t$, $y(0) = 0$, $y'(0) = 0$

[KTU Apr 2018]

Solution

$$[s^2 L(y) - sy(0) - y'(0)] + L(y) = 3L(\cos 2t)$$

$$(s^2 + 1)L(y) = \frac{3s}{s^2 + 4}$$

$$L(y) = \frac{3s}{(s^2 + 4)(s^2 + 1)}$$

$$\frac{3s}{(s^2 + 4)(s^2 + 1)} = \frac{As + B}{s^2 + 4} + \frac{Cs + D}{s^2 + 1}$$

$$3s = (As + B)(s^2 + 1) + (Cs + D)(s^2 + 4)$$

$$A = -1, B = 0, C = 1, D = 0$$

$$L(y) = -\frac{s}{s^2 + 4} + \frac{s}{s^2 + 1}$$

$$y = L^{-1}\left[\frac{s}{s^2 + 1}\right] - L^{-1}\left[\frac{s}{s^2 + 4}\right]$$

$$y = \cos t - \cos 2t$$

9. Solve $y'' - 4y' + 5y = 4e^{3t}$, $y(0) = 2$, $y'(0) = 7$

[KTU May 2019]

Solution

$$[s^2L(y) - sy(0) - y'(0)] - 4[sL(y) - y(0)] + 5L(y) = 4L(e^{3t})$$

$$(s^2 - 4s + 5)L(y) - 2s + 1 = 4 \cdot \frac{1}{s - 3}$$

$$(s^2 - 4s + 5)L(y) = \frac{4}{s - 3} + 2s - 1$$

$$L(y) = \frac{2s^2 - 7s + 7}{(s - 3)(s^2 - 4s + 5)}$$

$$\frac{2s^2 - 7s + 7}{(s - 3)(s^2 - 4s + 5)} = \frac{A}{(s - 3)} + \frac{Bs + C}{(s^2 - 4s + 5)}$$

$$2s^2 - 7s + 7 = (A)(s^2 - 4s + 5) + (Bs + C)(s - 3)$$

$$A = 2, B = 0, C = 1$$

$$\mathbf{L}(y) = 2\mathbf{L}^{-1}\left[\frac{1}{s-3}\right] + \mathbf{L}^{-1}\left[\frac{1}{s^2-4s+5}\right]$$

$$y = 2e^{-3t} + e^{2t} \sin t$$

4.8 LAPLACE TRANSFORM OF INTEGRAL

Let $F(s)$ denote the transform of a function $f(t)$ which is piecewise continuous for $t > 0$. Then for $s > 0$ and $t > 0$.

$$L\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s} F(s)$$

$$\int_0^t f(\tau) d\tau = L^{-1}\left\{\frac{1}{s} F(s)\right\}$$

4.9 SOLVED PROBLEMS

I Find the inverse transform of the following.

$$(1) \quad \frac{2}{s^2 + \frac{s}{3}}$$

Solution

$$\frac{2}{s^2 + \frac{s}{3}} = \frac{2}{s(s + \frac{1}{3})}$$

$$L^{-1}\left(\frac{1}{s + \frac{1}{3}}\right) = e^{-\frac{1}{3}t}$$

$$L^{-1}\left[\frac{1}{s(s + 1/3)}\right] = \int_0^t e^{-\frac{1}{3}\tau} d\tau = \left[\frac{e^{-\tau/3}}{-1/3}\right]_0^t$$

$$= -3\left[e^{-\frac{t}{3}} - 1\right] = 3\left(1 - e^{-\frac{t}{3}}\right)$$

$$L^{-1}\left(\frac{2}{s^2 + s/3}\right) = 6\left(1 - e^{-\frac{t}{3}}\right)$$

2. $\frac{1}{s\left(s^2 + \frac{w^2}{4}\right)}$

Solution

$$L^{-1}\left(\frac{1}{s^2 + w^2/4}\right) = \frac{2}{w} \sin \frac{w}{2} t$$

$$L^{-1}\left(\frac{1}{s\left(s^2 + w^2/4\right)}\right) = \int_0^t \frac{2}{w} \sin \frac{w}{2} \tau d\tau$$

$$= \frac{2}{w} \left[\frac{-\cos \frac{w}{2} \tau}{\frac{w}{2}} \right]_0^t = \frac{4}{w^2} \left(1 - \cos \frac{w}{2} t\right)$$

3. $\frac{s+8}{s^4 + 4s^2}$

Solution

$$\frac{s+8}{s^4 + 4s^2} = \frac{s+8}{s^2(s^2+4)}$$

$$L^{-1}\left(\frac{s+8}{s^2+4}\right) = L^{-1}\left(\frac{s}{s^2+4}\right) + 8L^{-1}\left(\frac{1}{s^2+4}\right)$$

$$= \cos 2t + 4 \sin 2t$$

$$L^{-1} \left(\frac{s+8}{s(s^2+4)} \right) = \int_0^t \cos 2\tau + 4 \sin 2\tau d\tau$$

$$= \left[\frac{\sin 2\tau}{2} - \frac{4 \cos 2\tau}{2} \right]_0^t$$

$$= \frac{\sin 2t}{2} - 2 \cos 2t + 2$$

$$L^{-1} \left(\frac{s+8}{s(s^2+4)} \right) = \int_0^t \left(\frac{\sin 2\tau}{2} - 2 \cos 2\tau + 2 \right) d\tau$$

$$= \left[-\frac{\cos 2\tau}{4} - \sin 2\tau + 2\tau \right]_0^t$$

$$= -\frac{\cos 2t}{4} - \sin 2t + 2t + \frac{1}{4}$$

4. $\frac{1}{s^3 + as^2}$

Solution

$$\frac{1}{s^3 + as^2} = \frac{1}{s^2(s+a)}$$

$$L^{-1} \left(\frac{1}{s+a} \right) = e^{-at}$$

$$L^{-1} \left(\frac{1}{s(s+a)} \right) = \int_0^t e^{-a\tau} d\tau = \left(\frac{e^{-a\tau}}{-a} \right)_0^t = \frac{1 - e^{-at}}{a}$$

$$L^{-1} \left(\frac{1}{s^2(s+a)} \right) = \int_0^t \frac{1 - e^{-a\tau}}{a} d\tau$$

$$\begin{aligned}
 &= \frac{1}{a} \left[\tau - \frac{e^{-a\tau}}{-a} \right]_0^1 = \frac{1}{a} \left[t + \frac{e^{-at}}{a} - \frac{1}{a} \right] \\
 &= \frac{1}{a^2} (e^{-at} - 1) + \frac{t}{a}
 \end{aligned}$$

EXERCISES

I. Solve the following IVP using Laplace transform

- (1) $\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 5y = e^{-t} \sin t, y(0) = 0, y'(0) = 1$
- (2) $y'' + 3y' + 2y = 8 \cos 2t, y(0) = -1, y'(0) = 2$
- (3) $y'' + 2y' + 6y = 6te^{-t}, y(0) = 2, y'(0) = 5$
- (4) $y'' - 3y' + 2y = 4, y(0) = 2, y'(0) = 3$
- (5) $y'' - 2y' - 8y = 0, y(0) = 3, y'(0) = 6$
- (6) $y'' + n^2 y = a \sin(nt + 2), y(0) = 0, y'(0) = 0$
- (7) $y''' - 3y'' + 3y' - y = t^2 e^t, y(0) = 1, y'(0) = 0, y''(0) = -2$
- (8) $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 3y = e^{-t}, y(0) = y'(0) = 1$
- (9) $y'' + 2y' + 2y = 5 \sin t, y(0) = 0, y'(0) = 0$

II Find the inverse transform of the following

- (1) $\frac{s^2 + 3}{s(s^2 + 9)}$
- (2) $\frac{1}{s^2(s+1)}$
- (3) $\frac{1}{s(s^2 - 16)}$
- (4) $\frac{1}{(s^3 - s)}$
- (4) $\frac{9}{s^2} \frac{s+1}{(s^2 + 9)}$

ANSWERS**I.**

$$(1) \quad \frac{1}{3} [e^{-t} \sin t + e^{-t} \sin 2t]$$

$$(2) \quad -\frac{2}{5} \cos 2t + \frac{12}{5} \frac{\sin 2t}{2} - \frac{8}{5} e^{-t} + e^{-2t}$$

$$(3) \quad \frac{e^{-t}}{5} \left[6t + 10 \cos \sqrt{5}t + \frac{29}{\sqrt{5}} \sin \sqrt{5}t \right]$$

$$(4) \quad 2 - 3e^t + 3e^{2t}$$

$$(5) \quad 2e^{4t} + e^{-2t}$$

$$(6) \quad \frac{1}{2n^3} (-nt \cos nt + \sin nt)$$

$$(7) \quad e^t - te^t - \frac{t^2 e^t}{2} + \frac{t^5 e^t}{60}$$

$$(8) \quad \frac{1}{4} (7e^{-t} - 3e^{-t} - 2te^{-t})$$

$$(9) \quad 2e^{-t} \cos t + e^{-t} \sin t - 2 \cos t + \sin t$$

II

$$(1) \quad \frac{1}{3} [1 + 2 \cos 3t]$$

$$(2) \quad t - 1 + e^{-t}$$

$$(3) \quad \frac{1}{16} (\cosh 4t - 1)$$

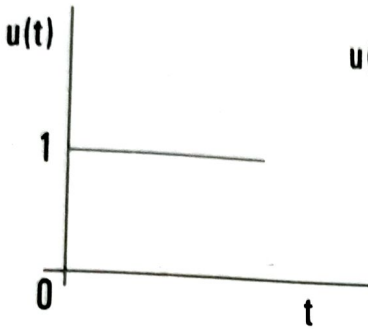
$$(4) \quad \cosh t - 1$$

$$(4) \quad 1 + t - \cos 3t - \frac{1}{3} \sin 3t$$

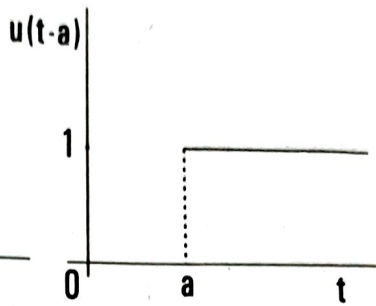
4.10 UNIT STEP FUNCTION (HEAVISIDE FUNCTION)

The unit step function or Heaviside function $U(t-a)$ is 0 for $t < a$, has a jump of size 1 at $t=a$ and is 1 for $t > a$

$$u(t-a) = \begin{cases} 0, & \text{if } t < a \\ 1, & \text{if } t > a \end{cases}$$



Unit step function $u(t)$



Unit step function $u(t-a)$

The Laplace transform of unit step function is

$$L(u(t-a)) = \int_0^{\infty} e^{-st} u(t-a) dt = \int_a^{\infty} e^{-st} 1 dt = \left[\frac{e^{-st}}{-s} \right]_a^{\infty} = \frac{e^{-as}}{s}$$

4.11 SECOND SHIFTING THEOREM (t SHIFTING)

If $f(t)$ has the transform $F(s)$ then the shifted function

$$\bar{f}(t) = f(t-a)u(t-a) = \begin{cases} 0, & \text{if } t < a \\ f(t-a), & \text{if } t > a \end{cases}$$

has the transform

$$e^{-as}F(s)$$

If $L[f(t)] = F(s)$ then

$$L[f(t-a)u(t-a)] = e^{-as}F(s)$$

Take the inverse on both sides,

$$f(t-a)u(t-a) = L^{-1}(e^{-as}F(s))$$

Also if $L(f(t)) = F(s)$, then

$$L[f(t)u(t-a)] = e^{-as}L(f(t+a))$$

4.12 SOLVED PROBLEMS

I Represent the following functions using unit step function and find its transform

(1) $t-3 \ (t>3)$

Solution

$$f(t) = (t-3) u(t-3)$$

$$L(f(t)) = L((t-3)u(t-3)) = e^{-3s} L(t) = e^{-3s} \frac{1}{s^2}$$

2) $e^{-t} \ (0 < t < \pi)$

Solution

$$f(t) = e^{-t} (1 - u(t - \pi))$$

$$= e^{-t} (1 - u(t - \pi))$$

$$= e^{-t} - e^{-t} u(t - \pi)$$

$$L[f(t)] = L(e^{-t}) - L(e^{-t} u(t - \pi))$$

$$= \frac{1}{s+1} - e^{-\pi s} L(e^{-t-\pi}) = \frac{1}{s+1} - e^{-\pi s} e^{-\pi} L(e^{-t})$$

$$= \frac{1}{s+1} - e^{-\pi s} e^{-\pi} L(e^{-t}) = \frac{1}{s+1} - e^{-\pi(s+1)} \frac{1}{s+1}$$

$$= \frac{1 - e^{-\pi(s+1)}}{s+1}$$

3) $2t^2 \left(t > \frac{5}{2} \right)$

Solution

$$f(t) = 2t^2 \left(u\left(t - \frac{5}{2}\right) \right)$$

$$L(f(t)) = 2L\left[t^2 u\left(t - \frac{5}{2}\right)\right]$$

$$= 2e^{-\frac{5}{2}s} L\left[\left(t + \frac{5}{2}\right)^2\right] = 2e^{-\frac{5}{2}s} L\left[t^2 + 5t + \frac{25}{4}\right]$$

$$= 2e^{-\frac{5}{2}s} \left(\frac{2}{s^3} + \frac{5}{s^2} + \frac{25}{4s} \right) = \frac{e^{-\frac{5}{2}s}}{2s^3} (25s^2 + 20s + 8)$$

4. $\cos 2t (0 < t < \pi)$

Solution

$$f(t) = \cos 2t (1 - u(t - \pi)) = \cos 2t - \cos 2t u(t - \pi)$$

$$L(f(t)) = L(\cos 2t) - L(\cos 2t u(t - \pi))$$

$$= \frac{s}{s^2 + 4} - e^{-\pi s} L[\cos 2(t + \pi)]$$

$$= \frac{s}{s^2 + 4} - e^{-\pi s} \frac{1}{s^2 + 4} = \frac{s - e^{-\pi s}}{s^2 + 4}$$

5. $e^{\frac{-\pi}{2}t} (1 < t < 3)$

Solution

$$f(t) = e^{\frac{-\pi}{2}t} (u(t - 1) - u(t - 3))$$

$$= e^{\frac{-\pi}{2}t} u(t - 1) - e^{\frac{-\pi}{2}t} u(t - 3)$$

$$L(f(t)) = L\left(e^{\frac{-\pi}{2}t} u(t - 1)\right) - L\left(e^{\frac{-\pi}{2}t} u(t - 3)\right)$$

$$= e^{-s} L\left(e^{\frac{\pi}{2}(t+1)}\right) - e^{-3s} L\left(e^{\frac{\pi}{2}(t+3)}\right)$$

$$= e^{-s} e^{-\pi/2} \frac{1}{s + \pi/2} - e^{-3s} e^{-3\pi/2} \frac{1}{s + \pi/2}$$

$$= \frac{e^{-s} e^{-\pi/2} - e^{-3s} e^{-3\pi/2}}{s + \pi/2} = \frac{e^{-s-\pi/2} - e^{-3s-3\pi/2}}{s + \pi/2}$$

6. $\sin t \left(\frac{\pi}{2} < t < \pi \right)$

Solution

$$f(t) = \sin t \left(u\left(t - \frac{\pi}{2}\right) - u(t - \pi) \right) = \sin t u\left(t - \frac{\pi}{2}\right) - \sin t u(t - \pi)$$

$$\begin{aligned}
L(f(t)) &= L\left(\sin tu\left(t - \frac{\pi}{2}\right)\right) - L(\sin tu(t - \pi)) \\
&= e^{-\frac{\pi}{2}s} L\left(\sin\left(t + \frac{\pi}{2}\right)\right) - e^{-\pi s} L(\sin(t + \pi)) \\
&= e^{-\frac{\pi}{2}s} L(\cos t) - e^{-\pi s} L(-\sin t) \\
&= e^{-\frac{\pi}{2}s} \frac{s}{s^2 + 1} + e^{-\pi s} \frac{1}{s^2 + 1} = \frac{se^{-\frac{\pi}{2}s} + e^{-\pi s}}{s^2 + 1}
\end{aligned}$$

7.
$$F(t) = \begin{cases} 2, & \text{if } 0 < t < 1 \\ \frac{t^2}{2}, & \text{if } 1 < t < \frac{\pi}{2} \\ \cos t, & \text{if } t > \frac{\pi}{2} \end{cases}$$

Solution

$$\begin{aligned}
f(t) &= 2(1 - u(t-1)) + \frac{t^2}{2} \left(u(t-1) - u\left(t - \frac{\pi}{2}\right) \right) + \cos t u\left(t - \frac{\pi}{2}\right) \\
&= 2 - 2u(t-1) + \frac{t^2}{2} u(t-1) - \frac{t^2}{2} u\left(t - \frac{\pi}{2}\right) + \cos t u\left(t - \frac{\pi}{2}\right)
\end{aligned}$$

$$\begin{aligned}
L(f(t)) &= 2L(1) - 2L(u(t-1)) + L\left(\frac{t^2}{2} u(t-1)\right) \\
&\quad - L\left[\frac{t^2}{2} u\left(t - \frac{\pi}{2}\right)\right] + L\left[\cos t u\left(t - \frac{\pi}{2}\right)\right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{2}{s} - 2e^{-s} L(t) + \frac{1}{2} e^{-s} L(t+1)^2 - \frac{1}{2} e^{-\frac{\pi}{2}s} L\left(t + \frac{\pi}{2}\right)^2 \\
&\quad + e^{-\frac{\pi}{2}s} L\left(\cos\left(t + \frac{\pi}{2}\right)\right)
\end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{s} - \frac{2e^{-s}}{s} + \frac{e^{-s}}{2} \left(\frac{2}{s^3} + \frac{\pi}{s^2} + \frac{1}{s} \right) \\
 &\quad - \frac{1}{2} e^{-\frac{\pi s}{2}} \left(\frac{2}{s^3} + \frac{\pi}{s^2} + \frac{\pi^2}{4s} \right) + e^{-\frac{\pi s}{2}} \left(-\frac{1}{s^2 + 1} \right) \\
 &= \frac{2}{s} - \frac{2e^{-s}}{s} + e^{-s} \left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s} \right) \\
 &\quad - e^{-\frac{\pi s}{2}} \left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s} \right) - \frac{e^{-\frac{\pi s}{2}}}{s^2 + 1}
 \end{aligned}$$

II. Find the inverse Laplace transform of the following functions

(1) $\frac{e^{-2s}}{s^6}$

Solution

$$L^{-1} \left(\frac{1}{s^6} \right) = \frac{t^5}{5!}$$

$$L^{-1} \left(\frac{e^{-2s}}{s^6} \right) = \frac{(t-2)^5 u(t-2)}{5!}$$

(2) $4 \left(\frac{1 - e^{-\pi s}}{s^2 + 4} \right)$

Solution

$$L^{-1} \left[\frac{4(1 - e^{-\pi s})}{s^2 + 4} \right] = 4 \left[L^{-1} \left(\frac{1}{s^2 + 4} \right) - L^{-1} \left(\frac{e^{-\pi s}}{s^2 + 4} \right) \right]$$

$$= 4 \left(\frac{\sin 2t}{2} - \frac{\sin(2t - \pi) u(2t - \pi)}{2} \right)$$

$$= 2(\sin 2t - \sin 2t u(t - \pi))$$

$$= 2 \sin 2t [1 - u(t - \pi)] = 2 \sin 2t, 0 < t < \pi$$

$$3. \quad 2 \left(\frac{e^{-s} - e^{-3s}}{s^2 - 4} \right)$$

Solution

$$\begin{aligned} L^{-1} \left(\frac{2(e^{-s} - e^{-3s})}{s^2 - 4} \right) &= 2 \left[L^{-1} \left(\frac{e^{-s}}{s^2 - 4} \right) - L^{-1} \left(\frac{e^{-3s}}{s^2 - 4} \right) \right] \\ &= 2 \left(\frac{\sinh(2t-1)u(t-1)}{2} - \frac{\sinh(2t-3)u(t-3)}{2} \right) \\ &= \sinh(2t-1)u(t-1) - \sinh(2t-3)u(t-3) \end{aligned}$$

$$4. \quad \frac{e^{-s}}{s^2 + \pi^2} + \frac{e^{-2s}}{s^2 + \pi^2} + \frac{e^{-3s}}{(s+2)^2}$$

Solution

$$\begin{aligned} f(t) &= L^{-1} \left(\frac{e^{-s}}{s^2 + \pi^2} \right) + L^{-1} \left(\frac{e^{-2s}}{s^2 + \pi^2} \right) + L^{-1} \left(\frac{e^{-3s}}{(s+2)^2} \right) \\ &= \frac{\sin \pi(t-1)u(t-1)}{\pi} + \frac{\sin \pi(t-2)u(t-2)}{\pi} \\ &\quad + e^{-2(t-3)}(t-3)u(t-3) \end{aligned}$$

$$= \frac{-\sin \pi t u(t-1)}{\pi} + \frac{\sin \pi t u(t-2)}{\pi} + e^{-2(t-3)}(t-3)u(t-3)$$

$$f(t) = \begin{cases} 0, & 0 < t < 1 \\ \frac{-\sin \pi t}{\pi}, & 1 < t < 2 \\ 0, & 2 < t < 3 \\ e^{-2(t-3)}(t-3), & t > 3 \end{cases}$$

$$5. \quad \frac{(1 + e^{-2\pi(s+1)})(s+1)}{(s+1)^2 + 1}$$

$$\begin{aligned}
 f(t) &= L^{-1} \left[\frac{(s+1)}{(s+1)^2 + 1} \right] + e^{-2\pi} L^{-1} \left[\frac{e^{-2\pi s} (s+1)}{(s+1)^2 + 1} \right] \\
 &= e^{-t} \cos t + e^{-2\pi} e^{-(t-2\pi)} \cos(t-2\pi) u(t-2\pi) \\
 &= e^{-t} \cos t - e^{-t} \cos t u(t-2\pi) \\
 &= e^{-t} \cos t (1 - u(t-2\pi)) \\
 &= e^{-t} \cos t, \quad 0 < t < 2\pi
 \end{aligned}$$

$$6. \quad \frac{2s-10}{s^3} e^{-5s}$$

$$\begin{aligned}
 L^{-1} \left[\frac{2s-10}{s^3} e^{-5s} \right] &= 2L^{-1} \left[\frac{e^{-5s}s}{s^3} \right] - 10L^{-1} \left[\frac{e^{-5s}}{s^3} \right] \\
 &= 2L^{-1} \left[\frac{e^{-5s}}{s^2} \right] - 10L^{-1} \left[\frac{e^{-5s}}{s^3} \right] = 2(t-5)u(t-5) - 5(t-5)^2 u(t-5)
 \end{aligned}$$

$$7. \quad \frac{s-1}{s^2} e^{-s}$$

$$L^{-1} \left[\frac{s-1}{s^2} e^{-s} \right] = L^{-1} \left[\frac{e^{-s}}{s} \right] - L^{-1} \left[\frac{e^{-s}}{s^2} \right] = u(t-1) - (t-1)u(t-1)$$

EXERCISES

I. Express the following function in terms of unit step function and hence find its Laplace transform.

$$(a) \quad f(t) = \begin{cases} 2, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \\ \sin t, & t > 2\pi \end{cases}$$

$$(b) \quad f(t) = \begin{cases} 2+t^2, & 0 < t < 2 \\ 6, & 2 < t < 3 \\ \frac{2}{2t-5}, & t > 3 \end{cases}$$

2. Find the inverse Laplace transform of the following

$$(a) \quad \frac{e^{-\pi s}}{s^2 + 1}$$

$$(b) \quad \frac{se^{-s/2} + \pi e^{-s}}{s^2 + \pi^2}$$

$$(c) \quad \frac{3}{5} - \frac{4e^{-s}}{s^2} + \frac{4e^{-3s}}{s^2}$$

$$(d) \quad \frac{2}{s^2} - \frac{2e^{-2s}}{s^2} - \frac{4e^{-2s}}{s} + \frac{se^{-\pi s}}{s^2 + 1}$$

ANSWERS

I

$$(a) \quad \frac{2}{s} - \frac{2e^{-\pi s}}{s} + \frac{4e^{-2\pi s}}{s^2 + 1}$$

$$(b) \quad (2 + t^2)u(t) + (4 - t^2)u(t-2) + \frac{32-12t}{2t-5}u(t-3)$$

II

$$(a) \quad -\sin t u(t - \pi)$$

$$(b) \quad \sin \pi t \left[u\left(t - \frac{1}{2}\right) - u(t-1) \right]$$

$$(c) \quad 3 - 4u(t-1)u(t-1) + 4u(t-3)u(t-3)$$

$$(d) \quad 2t - 2u(t-2)u(t-2) - 4u(t-2) + \cos(t - \pi)u(t - \pi)$$

4.15 CONVOLUTION THEOREM

The transform of a product is generally different from the product of the transforms of the factors

$$L(fg) \neq L(f).L(g)$$

Take

$$f = e^t \text{ and } g = 1$$

$$fg = e^t$$

$$L(fg) = \frac{1}{s-1}$$

$$L(f) = \frac{1}{s-1} L(g) = \frac{1}{s}$$

$$L(f) \cdot L(g) = \frac{1}{s^2 - s}$$

$$L(fg) \neq L(f)L(g)$$

In Laplace transform, the transform of the convolution of f and g , denoted by the standard notation $f * g$ and define by the integral.

$$h(t) = (f * g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$$

Convolution theorem

If

$$L(f(t)) = F(s) \text{ and } L(g(t)) = G(s)$$

then

$$L[f(t) * g(t)] = F(s)G(s) \text{ and}$$

$$L^{-1}[F(s)G(s)] = f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)dt$$

4.16 SOLVED PROBLEMS

1. Find $1 * (-1)$

Solution

$$(1 * (-1))(t) = \int_0^t -1 d\tau = -t$$

2. Find $e^{-t} * e^t$

Solution

$$(e^{-t} * e^t)(t) = \int_0^t e^{-\tau} e^{t-\tau} d\tau = \int_0^t e^{t-2\tau} d\tau$$

$$= e^t \left[\frac{e^{-2\tau}}{-2} \right]_0^t = \frac{-e^t}{2} [e^{-2t} - 1]$$

$$= \frac{e^t - e^{-t}}{2} = \sinh t$$

3) $\cos wt * 1$

Solution

$$(\cos wt * 1)(t) = \int_0^t \cos w\tau d\tau = \left[\frac{\sin w\tau}{w} \right]_0^t = \frac{\sin wt}{w}$$

4. $t * e^{-t}$

Solution

$$\begin{aligned} (t * e^{-t})(t) &= \int_0^t \tau e^{-(t-\tau)} d\tau = e^{-t} \int_0^t \tau e^{\tau} d\tau \\ &= e^{-t} [\tau e^{\tau} - e^{\tau}]_0^t \\ &= e^{-t} [t e^t - e^t + 1] = t - 1 + e^{-t} \end{aligned}$$

5. $\cos wt * \cos wt$

Solution

$$\begin{aligned} (\cos wt * \cos wt)(t) &= \int_0^t \cos w\tau \cos w(t-\tau) d\tau \\ &= \frac{1}{2} \int_0^t \cos w\tau + \cos w(-t+2\tau) d\tau \\ &= \frac{1}{2} \left\{ \cos w\tau [\tau]_0^t + \left[\frac{\sin w(2\tau-t)}{2} \right]_0^t \right\} \\ &= \frac{1}{2} \left\{ t \cos wt + \frac{\sin wt}{2} + \frac{\sin wt}{2} \right\} \\ &= \frac{t}{2} \cos wt + \frac{\sin wt}{2} = \frac{t \cos wt + \sin wt}{2} \end{aligned}$$

6. $1 * \sin wt$

Solution

$$(1 * \sin wt)(t) = \int_0^t \sin w(t-\tau) d\tau = [\cos w(t-\tau)]_0^t = 1 - \cos wt$$

7. $e^{at} * e^{bt} \ (a \neq b)$

Solution

$$\begin{aligned} (e^{at} * e^{bt})(t) &= \int_0^t e^{a\tau} e^{b(t-\tau)} d\tau = e^{bt} \int_0^t e^{(a-b)\tau} d\tau \\ &= e^{bt} \left[\frac{e^{(a-b)\tau}}{a-b} \right]_0^t = \frac{e^{bt}}{a-b} (e^{(a-b)t} - 1) \\ &= \frac{e^{at} - e^{bt}}{a-b} \quad (a \neq b) \end{aligned}$$

8. If

$$L(f) = \frac{5.5}{(s+1.5)(s-4)}$$

Find inverse transform using convolution theorem

Solution

Let $F(s) = \frac{1}{s+1.5}, G(s) = \frac{1}{s-4}$

$$f(t) = L^{-1}\left(\frac{1}{s+1.5}\right) = e^{-1.5t}$$

$$g(t) = L^{-1}\left(\frac{1}{s-4}\right) = e^{4t}$$

$$L^{-1}[F(s)G(s)] = \int_0^t f(\tau)g(t-\tau) d\tau$$

$$\begin{aligned} L^{-1}\left(\frac{5.5}{(s+1.5)(s-4)}\right) \\ = 5.5 \int_0^t e^{-1.5\tau} e^{4(t-\tau)} d\tau = 5.5 e^{4t} \int_0^t e^{-5.5\tau} d\tau \end{aligned}$$

$$= 5.5 e^{4t} \left[\frac{e^{-5.5\tau}}{-5.5} \right]_0^t = -e^{4t} [e^{-5.5t} - 1]$$

$$= e^{4t} [1 - e^{-5.5t}] = e^{4t} - e^{-1.5t}$$

EXERCISES

Apply convolution theorem to find the inverse transform of the following

$$(1) \quad \frac{s}{(s^2 + a^2)^2}$$

$$(2) \quad \frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$$

$$(3) \quad \frac{s^2}{(s^2 + 4)^2}$$

$$(4) \quad \frac{1}{(s^2 + a^2)^2}$$

$$(5) \quad \frac{1}{s(s^2 - a^2)}$$

$$(6) \quad \frac{1}{(s+1)(s+9)^2}$$

$$(7) \quad \frac{3}{(s^2 + 1)(s^2 + 9)}$$

$$(8) \quad \frac{1}{(s+1)(s^2 + 1)}$$

ANSWERS

$$(1) \quad \frac{1}{2a} t \sin at$$

$$(2) \quad \frac{a \sin at - b \sin bt}{a^2 - b^2}$$

$$(3) \quad \frac{1}{2} \left[t \cos 2t + \frac{\sin 2t}{2} \right]$$

$$(4) \quad \frac{1}{2a^3} [\sin at - at \cos at]$$

$$(5) \quad \frac{1}{a^2} (\cosh at - 1)$$

$$(6) \quad \frac{e^{-t}}{64} (1 - e^{-8t} (1 + 8t))$$

$$(7) \quad \frac{1}{8} (3 \sin t - \sin 3t)$$

$$(8) \quad \frac{1}{2} [e^{-t} + \sin t - \cos t]$$